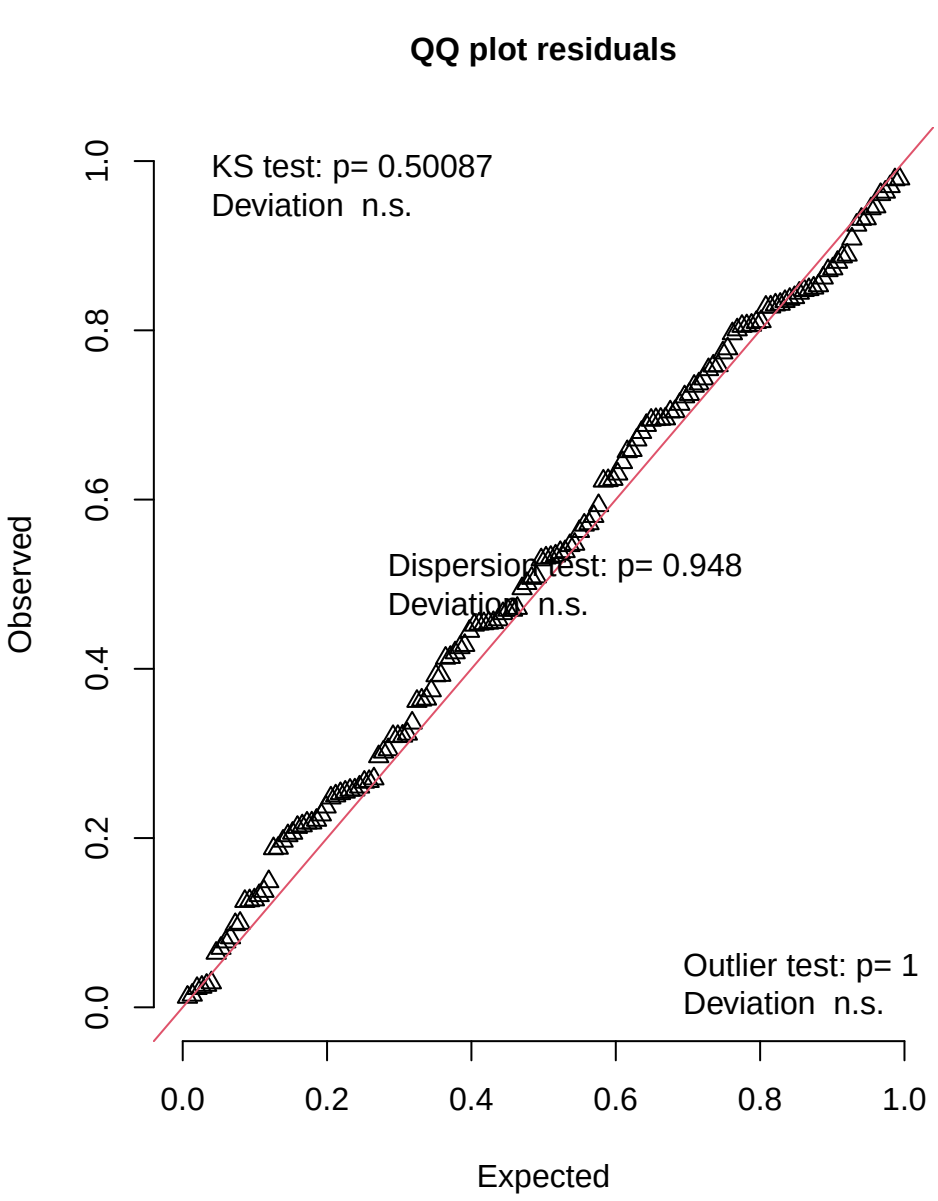
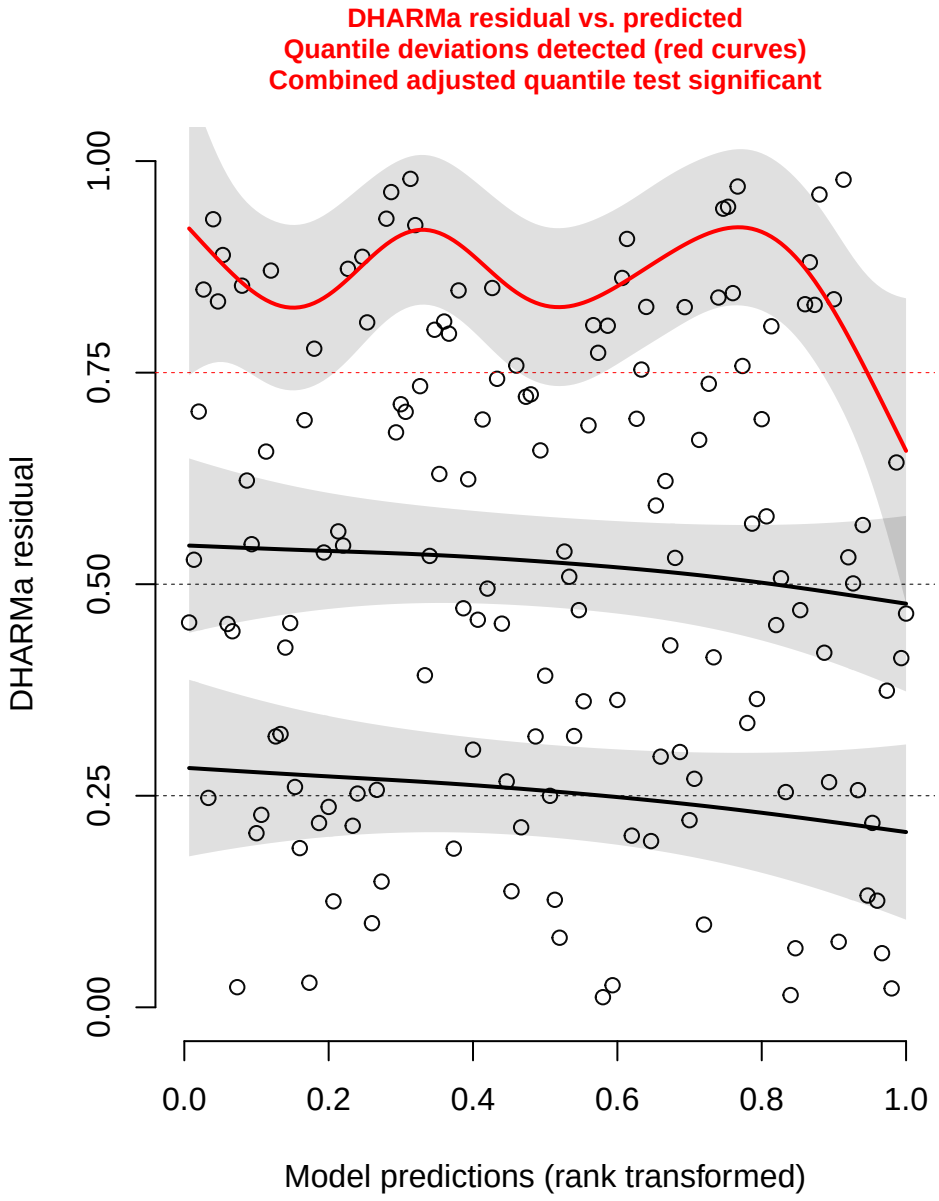
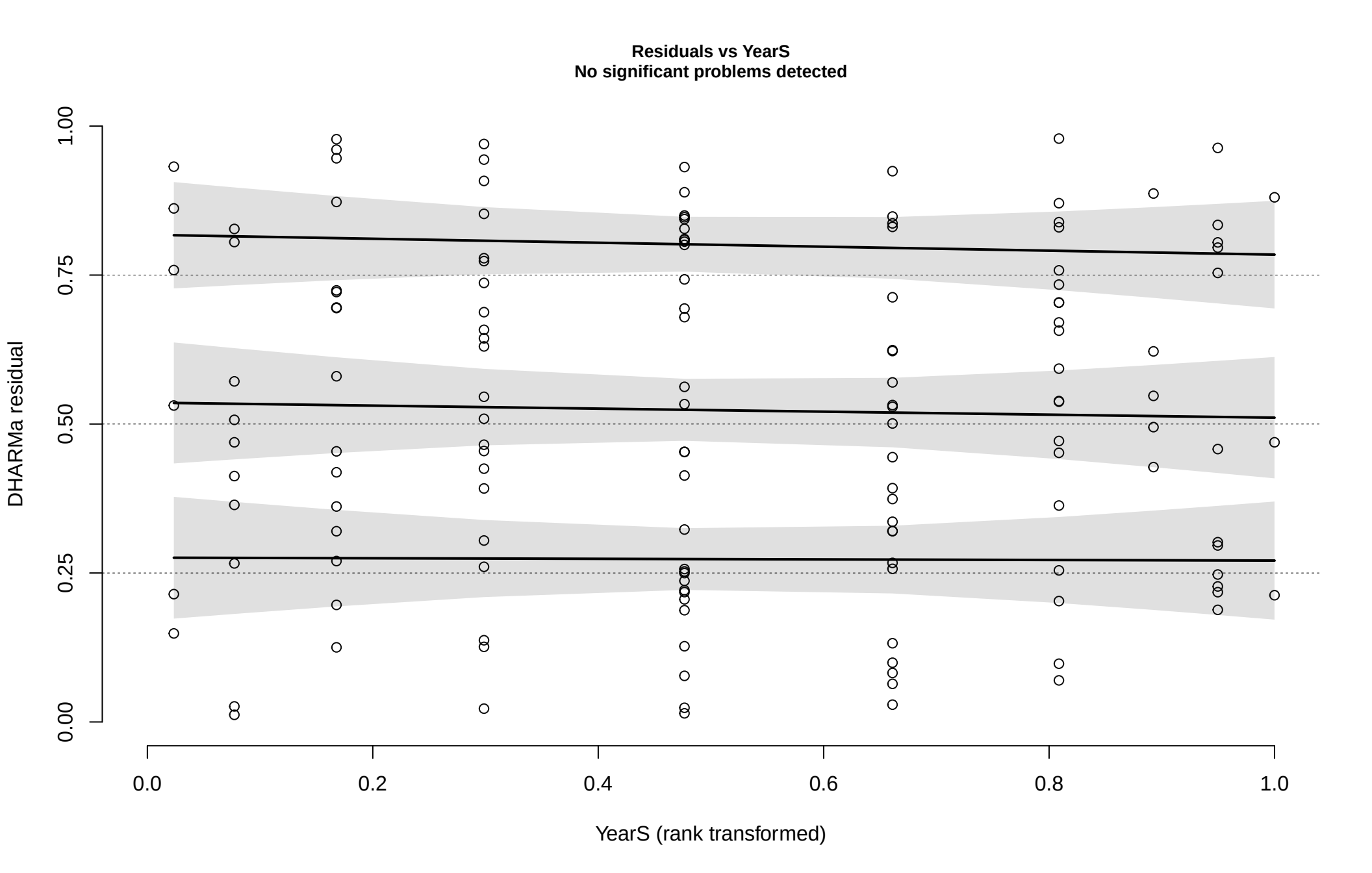
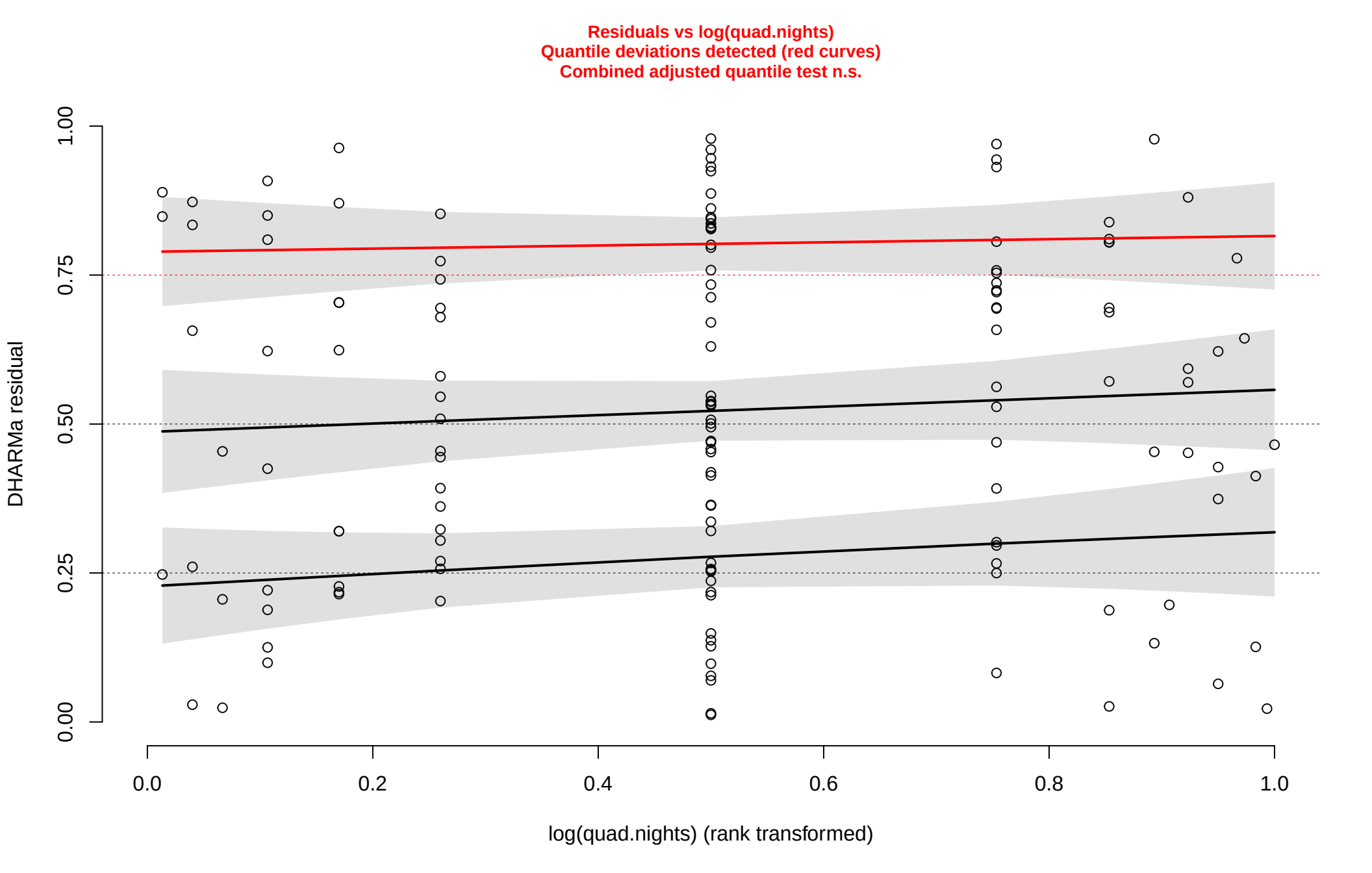
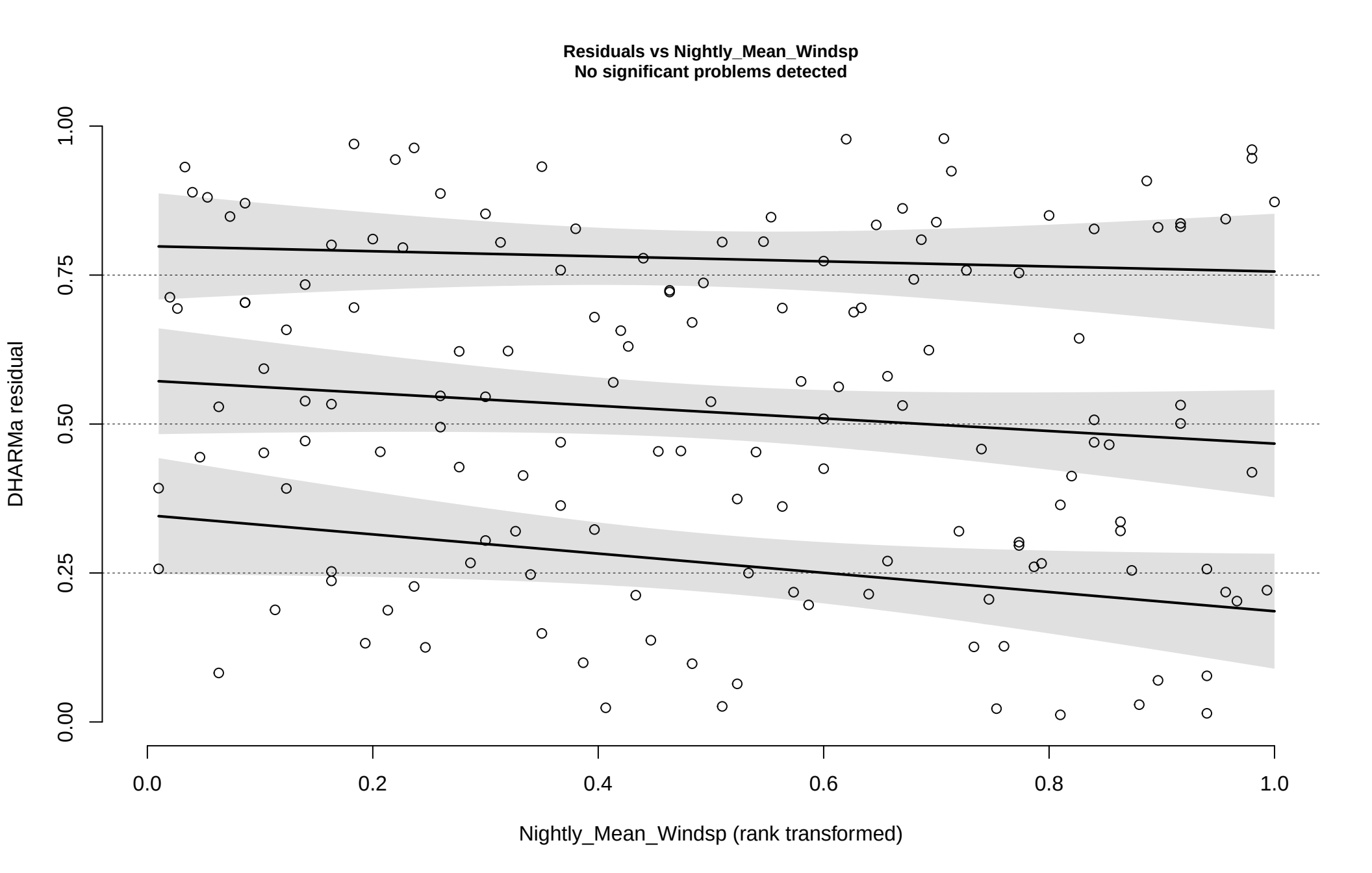
DHARMa residuals: Northern Region_EPFU

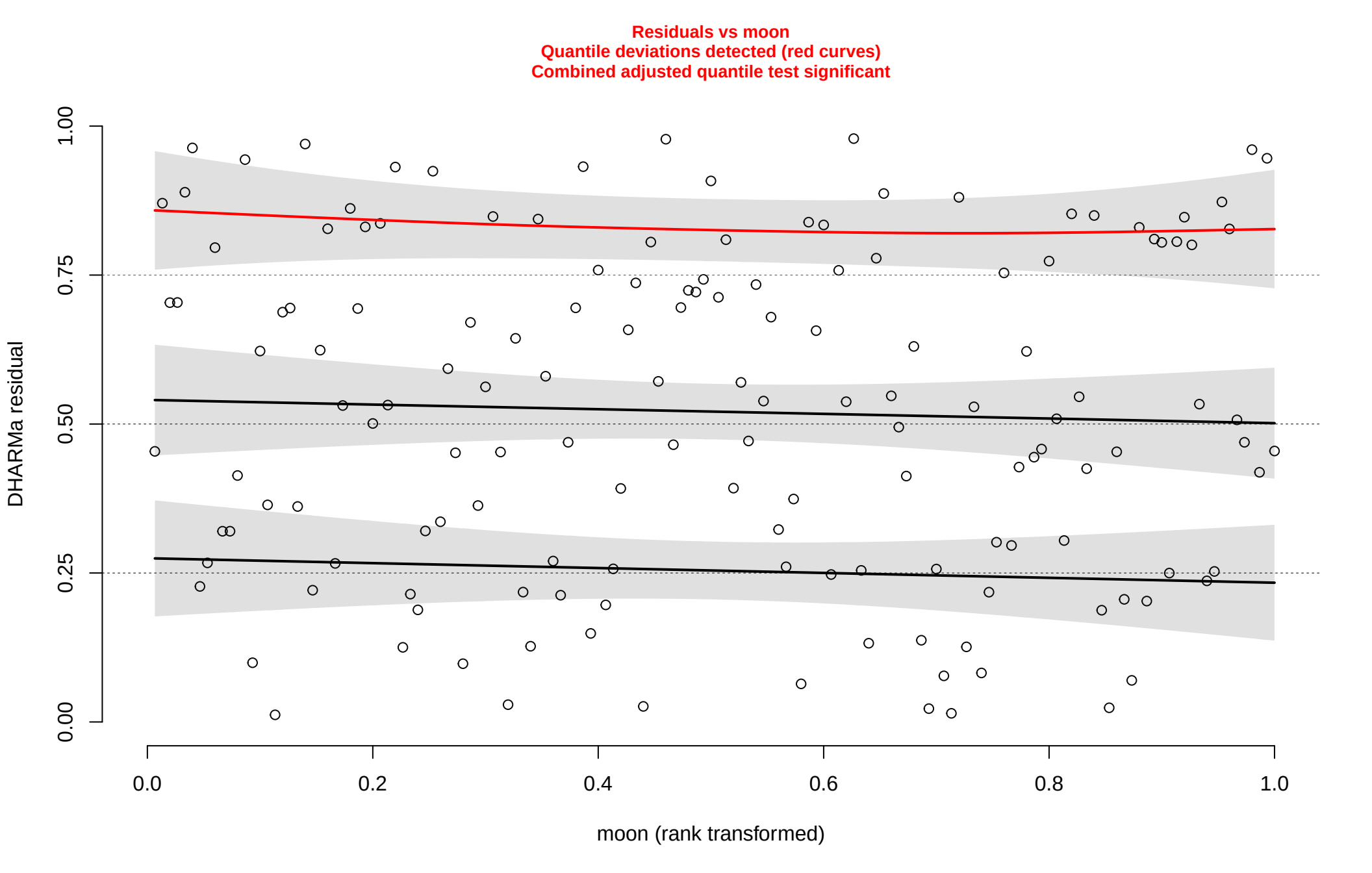


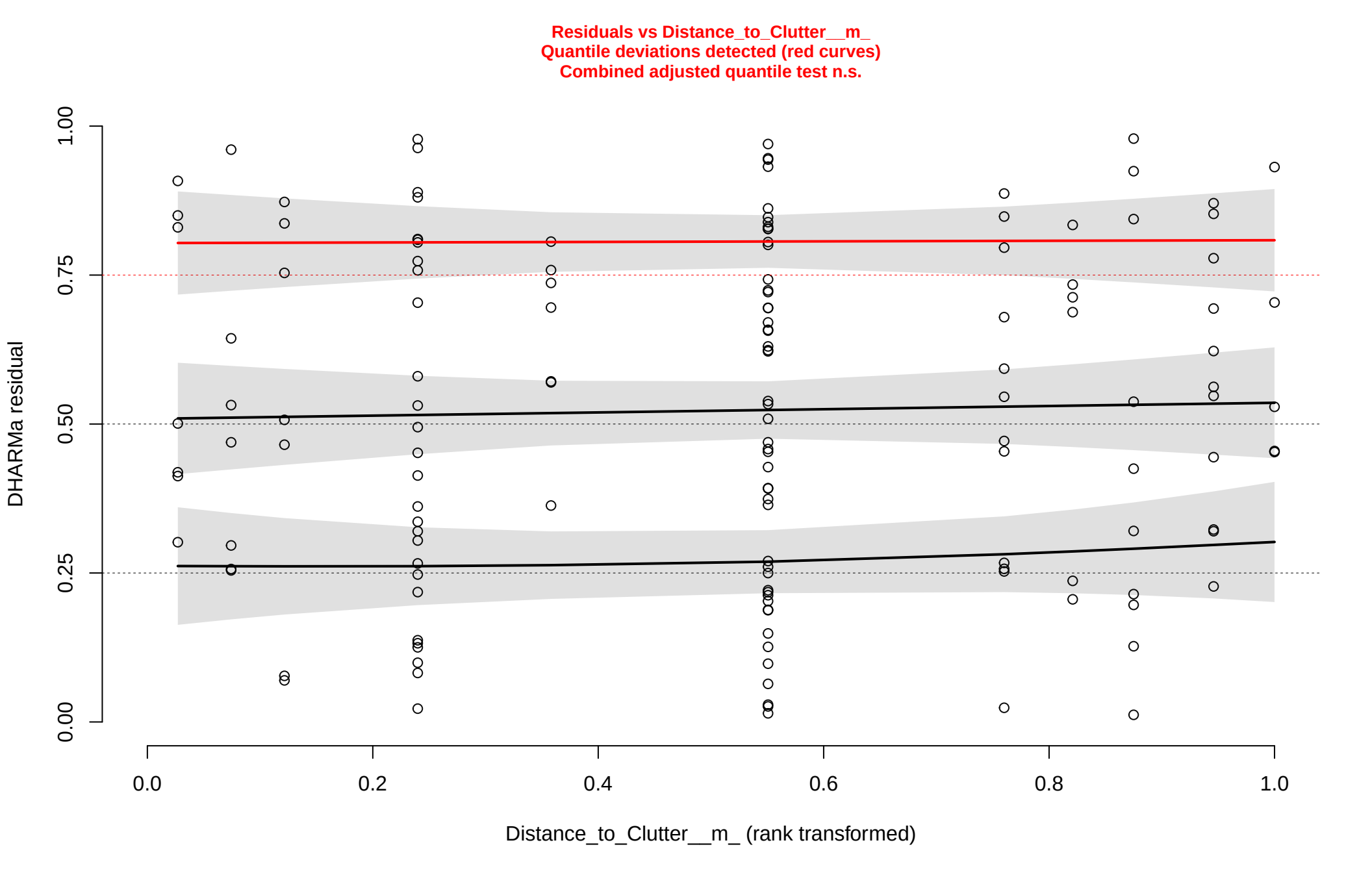


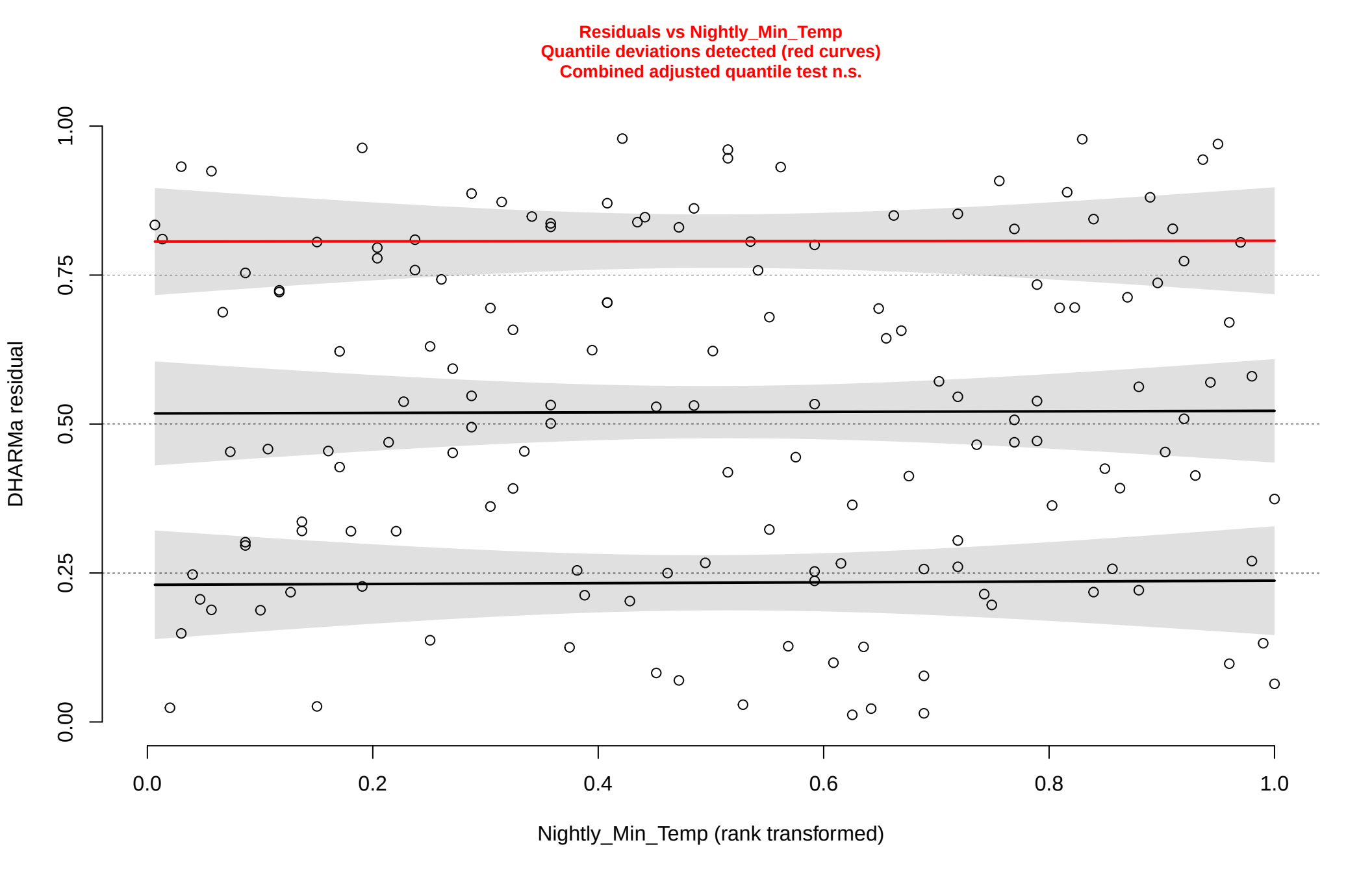

Residuals vs YearS
No significant problems detected
DHARMa residual
1.00
0.75
0.50
0.25
0.00
0.0
0.2
0.4
0.6
0.8
1.0
YearS (rank transformed)

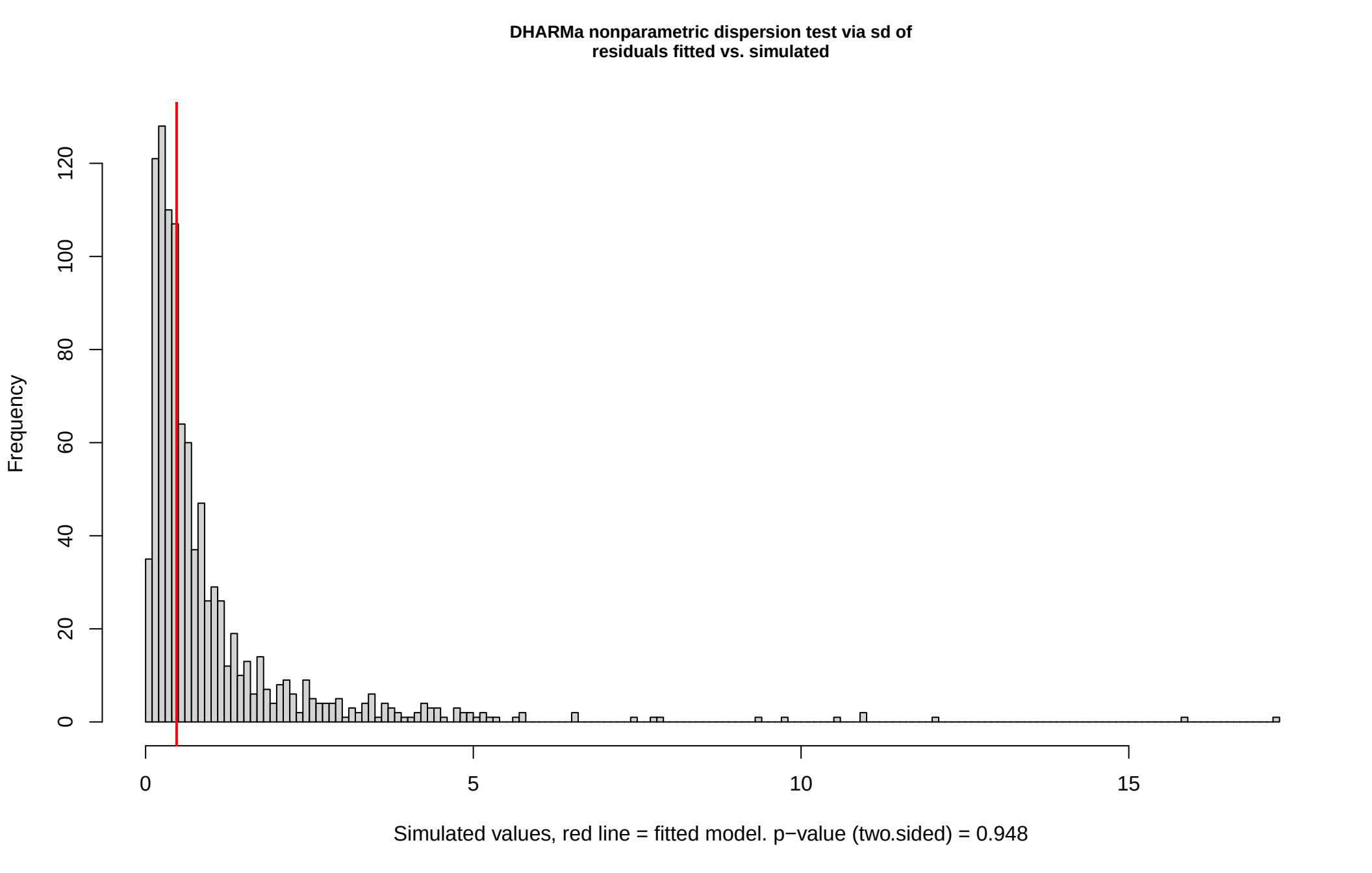

Residuals vs log(quad.nights)
Quantile deviations detected (red curves)
Combined adjusted quantile test n.s.
DHARMa residual
0.00
0.25
0.50
0.75
1.00
0.0
0.2
0.4
0.6
0.8
1.0
log(quad.nights) (rank transformed)

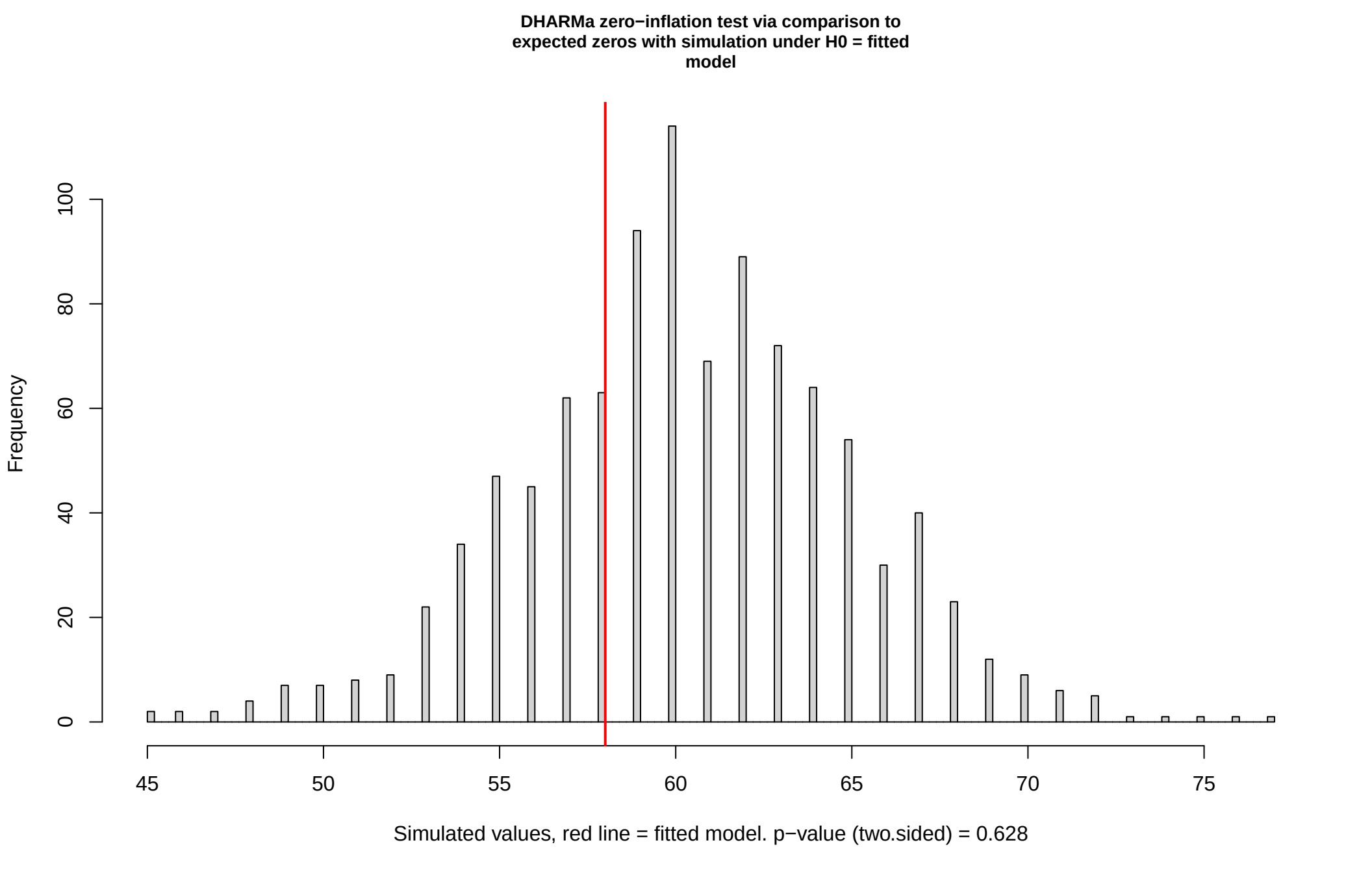

Residuals vs Nightly_Mean_Windsp
No significant problems detected
DHARMa residual
0.00
0.25
0.50
0.75
1.00
0.0
0.2
0.4
0.6
0.8
1.0
Nightly_Mean_Windsp (rank transformed)

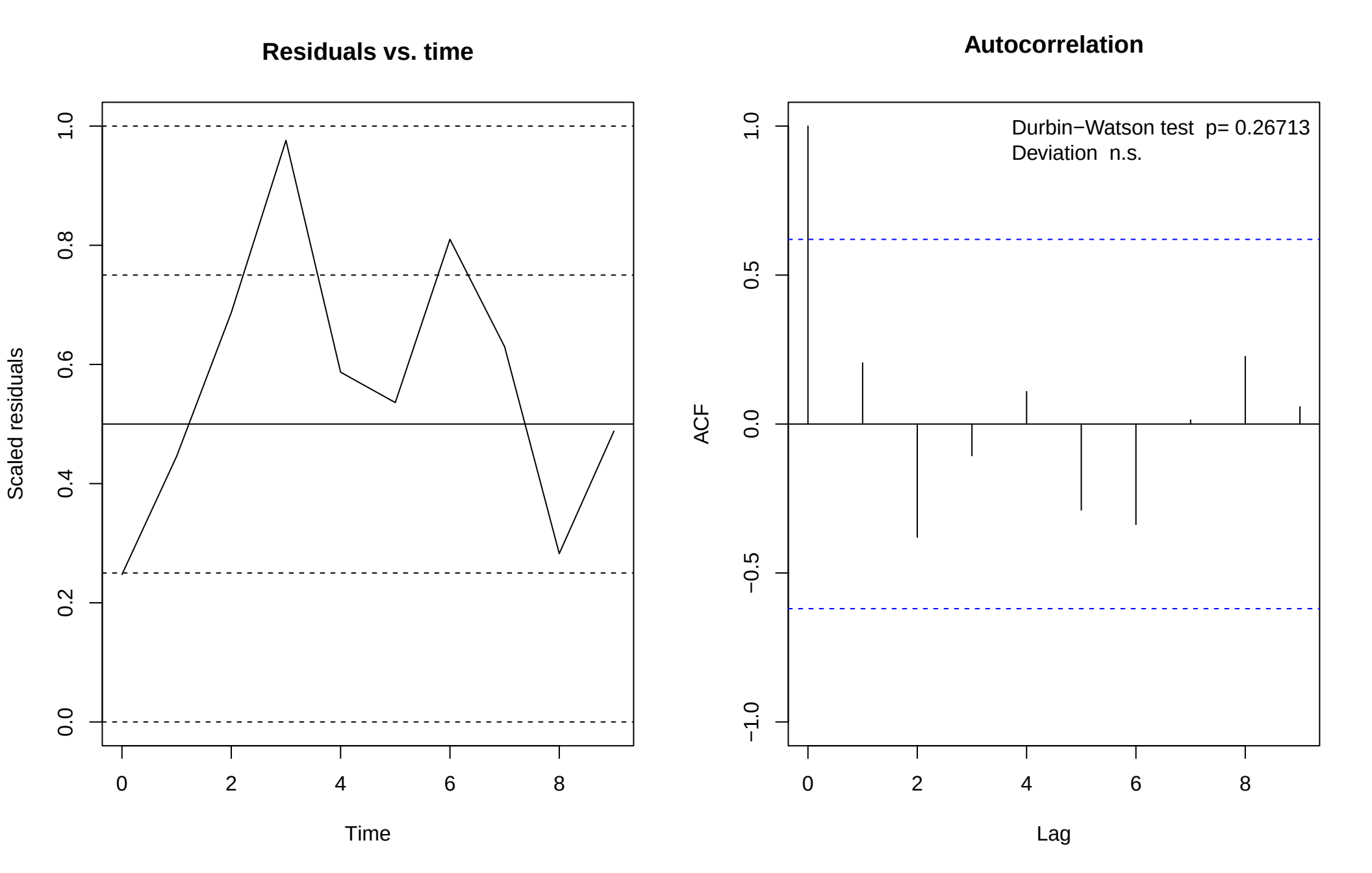

Residuals vs moon
Quantile deviations detected (red curves)
Combined adjusted quantile test significant
DHARMa residual
1.00
0.75
0.50
0.25
0.00
0.0
0.2
0.4
0.6
0.8
1.0
moon (rank transformed)

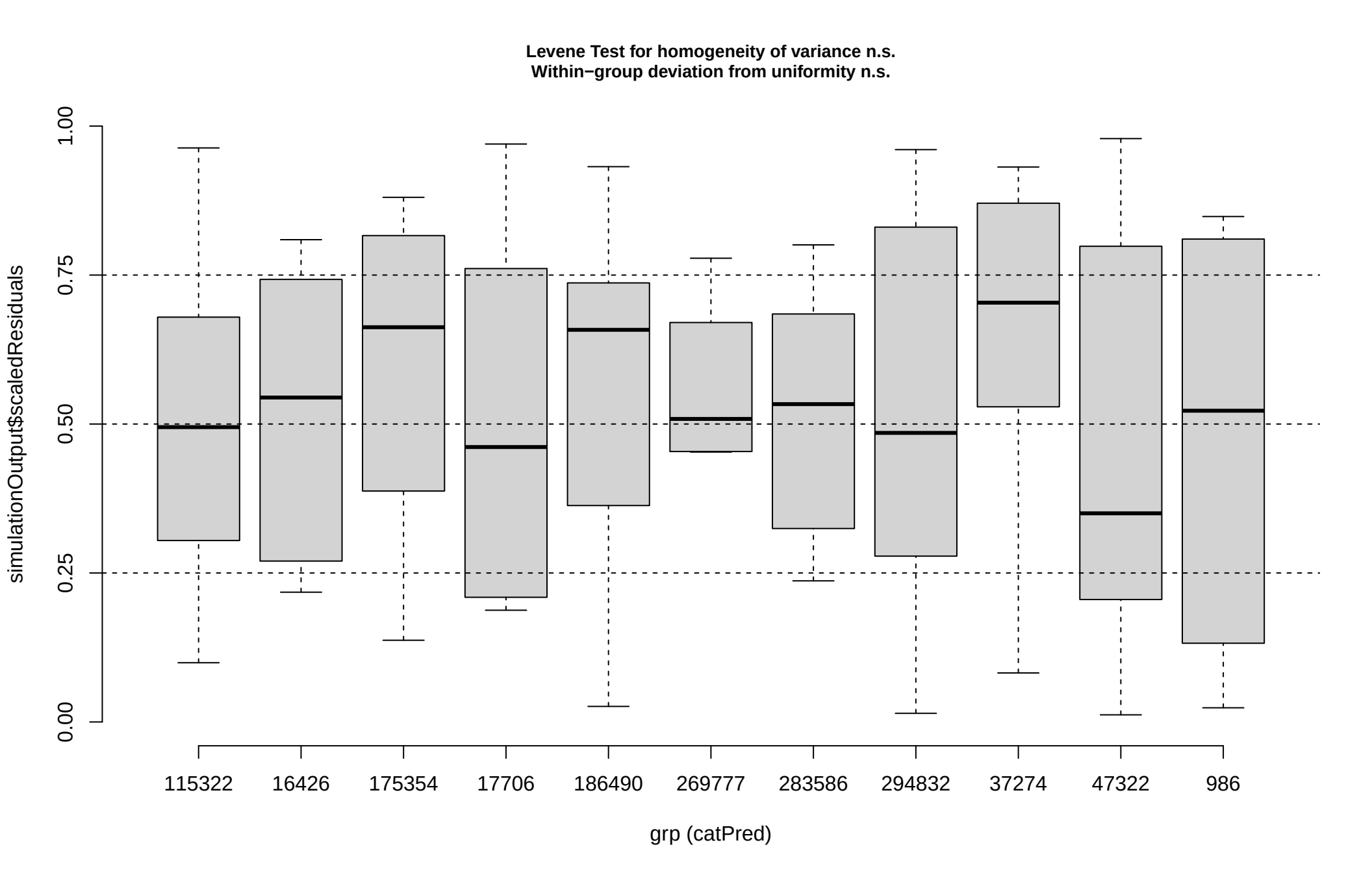

Residuals vs Distance_to_Clutter__m_
Quantile deviations detected (red curves)
Combined adjusted quantile test n.s.
DHARMa residual
1.00
0.75
0.50
0.25
0.00
0.0
0.2
0.4
0.6
0.8
1.0
Distance_to_Clutter__m_ (rank transformed)

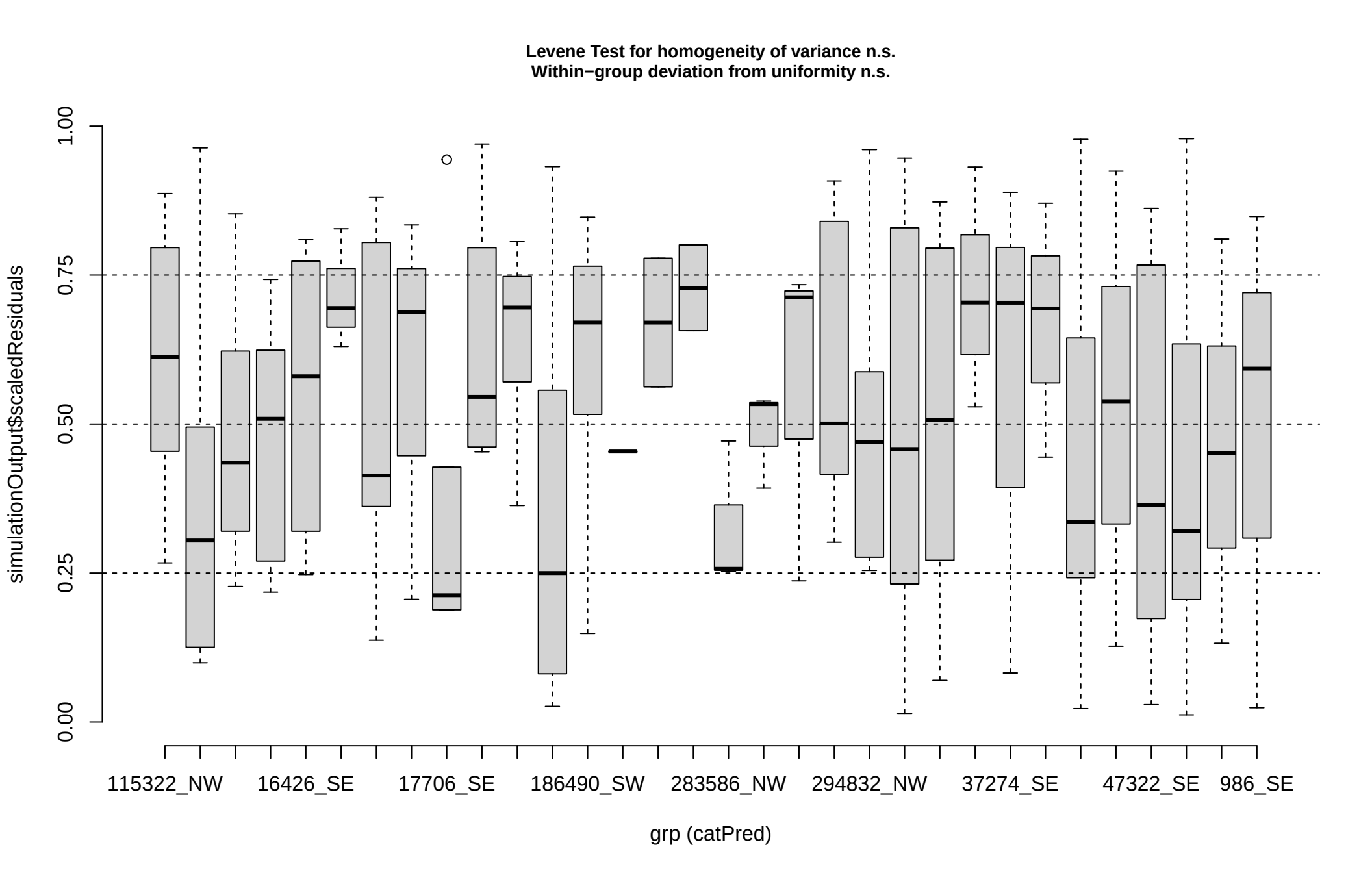

Residuals vs Nightly_Min_Temp
Quantile deviations detected (red curves)
Combined adjusted quantile test n.s.
DHARMa residual
1.00
0.75
0.50
0.25
0.00
0.0
0.2
0.4
0.6
0.8
1.0
Nightly_Min_Temp (rank transformed)

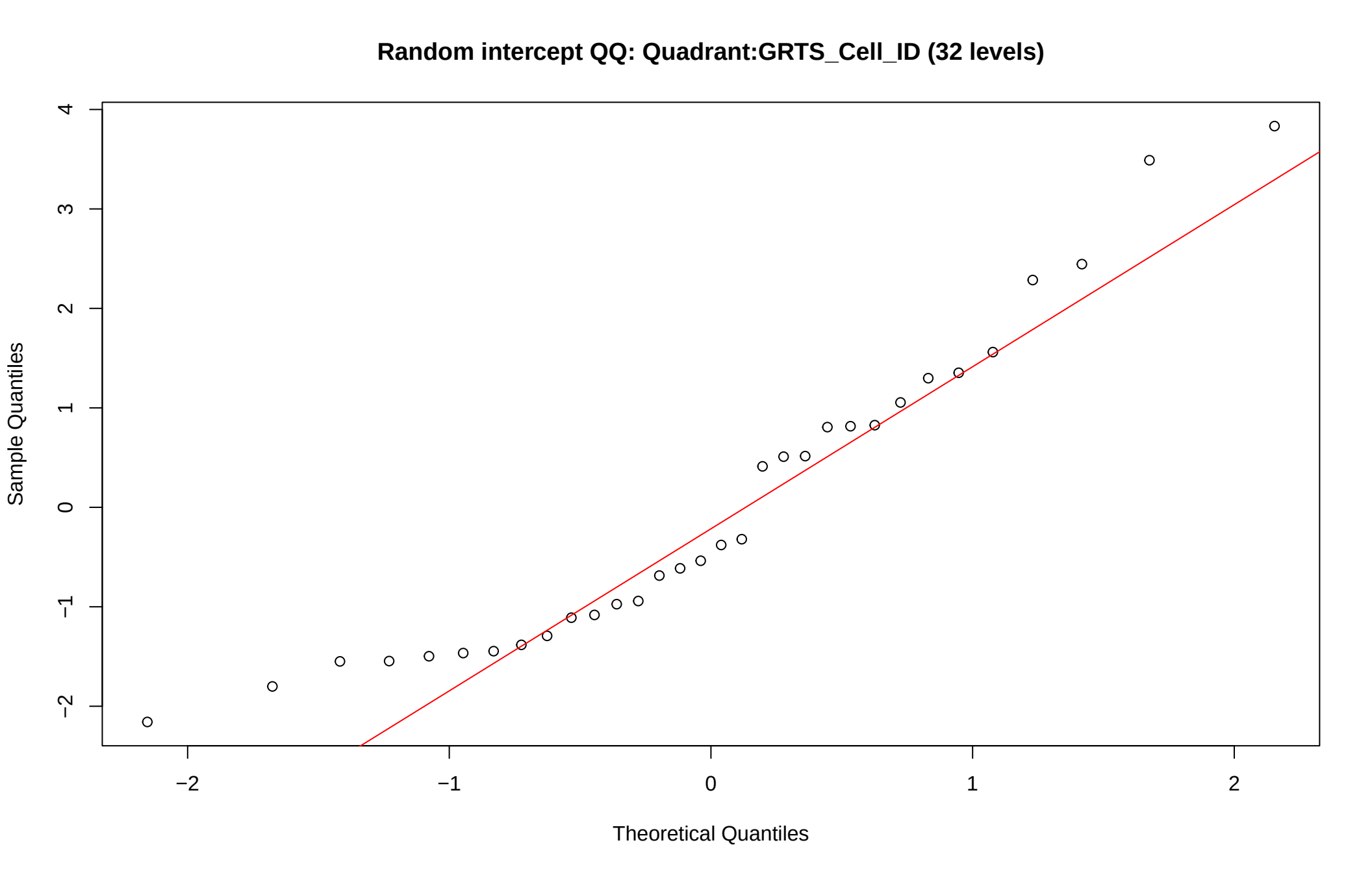

DHARMa nonparametric dispersion test via sd of
residuals fitted vs. simulated
Frequency
0
20
40
60
80
100
120
0
5
10
15
Simulated values, red line = fitted model. p−value (two.sided) = 0.948

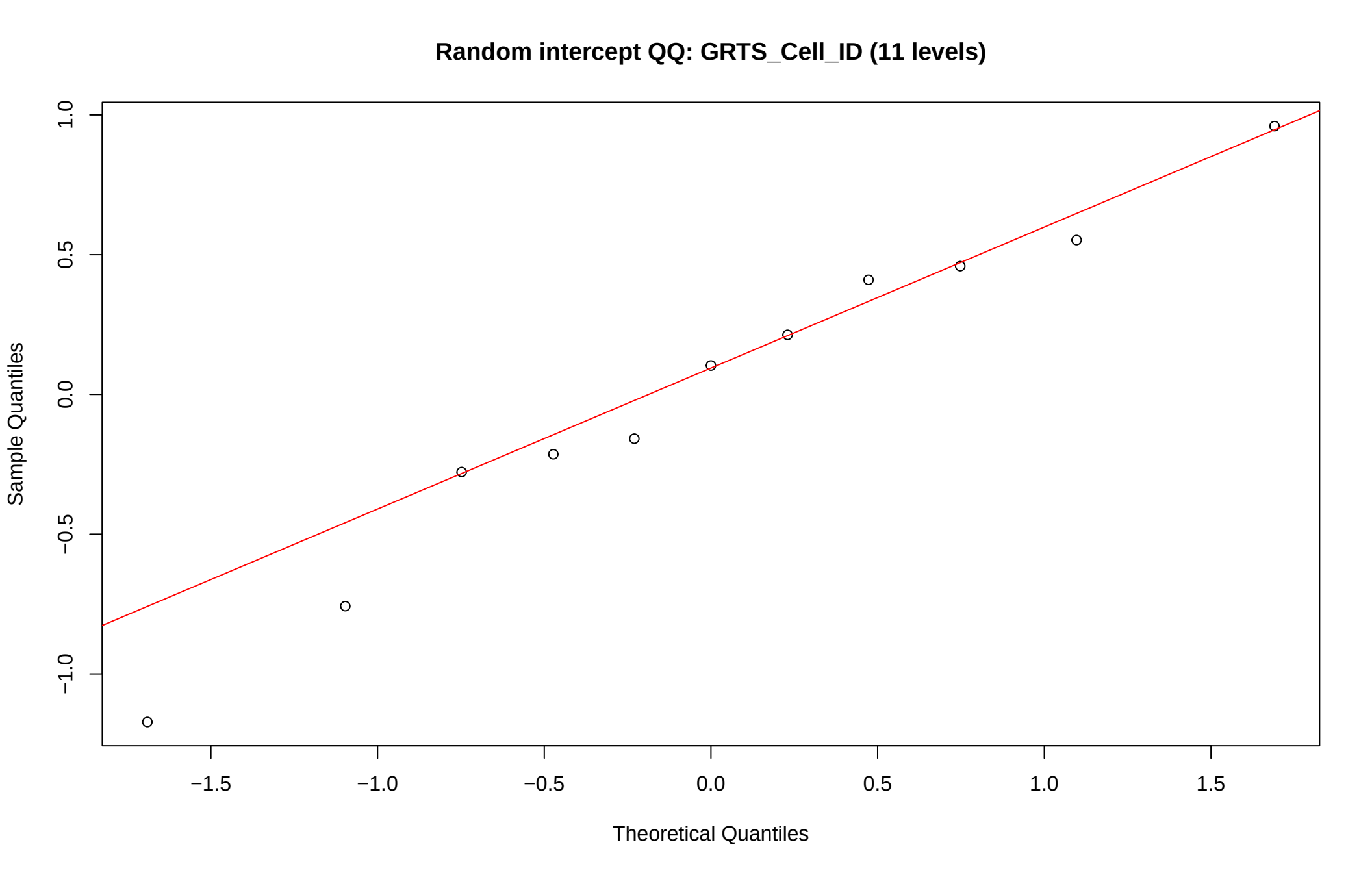

DHARMa zero−inflation test via comparison to expected zeros with simulation under H0 = fitted model
Frequency
0
20
40
60
80
100
45
50
55
60
65
70
75
Simulated values, red line = fitted model. p−value (two.sided) = 0.628


Residuals vs. time
Scaled residuals
1.0
0.8
0.6
0.4
0.2
0.0
0
2
4
6
8
Time
Autocorrelation
ACF
Durbin−Watson test p= 0.26713
Deviation n.s.
1.0
0.5
0.0
−0.5
−1.0
0
2
4
6
8
Lag


Levene Test for homogeneity of variance n.s.
Within−group deviation from uniformity n.s.
simulationOutput$scaledResiduals
1.00
0.75
0.50
0.25
0.00
115322
16426
175354
17706
186490
269777
283586
294832
37274
47322
986
grp (catPred)


Levene Test for homogeneity of variance n.s.
Within−group deviation from uniformity n.s.
simulationOutput$scaledResiduals
1.00
0.75
0.50
0.25
0.00
115322_NW
16426_SE
17706_SE
186490_SW
283586_NW
294832_NW
37274_SE
47322_SE
986_SE
grp (catPred)


Random intercept QQ: Quadrant:GRTS_Cell_ID (32 levels)
Sample Quantiles
Theoretical Quantiles
4
3
2
1
0
−1
−2
−2
−1
0
1
2


Random intercept QQ: GRTS_Cell_ID (11 levels)
Sample Quantiles
Theoretical Quantiles
1.0
0.5
0.0
−0.5
−1.0
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5